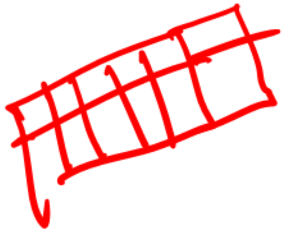
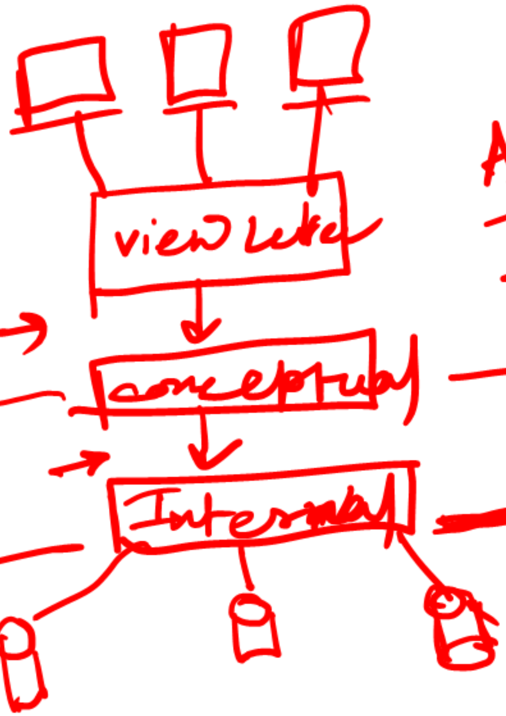
Data Independence.

→ Logical Data Independence

→ Physical " "





Application.

→ Structure Change.

Logical D.I.

Storage change/
Location Change

Physical D.I.

Relational Data Model:

Table.

Field/Attribute.

Table: Student → Entity set/Table/Relation.

sid	Name	Roll	Class	Sec	Phno	AdharNo	Address
S001	AA	2	12	B	9834...	7258----	Garia
S002	BB	10	11	C	6528...	2825---	Rubi
S003	CC	5	$\boxed{10}$	B	7278...	7505---	Jadavpur

Record/Tuple

Data.

Degree ⇒ Total no. of Field/Attribute ⇒ 8

Cardinality ⇒ Total No. of Record/Tuple ⇒ 3

RDBMS

Key

- Super Key.
- Candidate Key
- Primary Key
- Alternate Key / secondary key
- Composite Key
- Foreign Key.

→ Student {sid, name, Roll, class, sec, Phno, Adhno, Address}

Superkey :→ Uniquely Identify

{{sid}, {Phno.}, {Adhno}, {name, Roll, class, sec}, - - - - - - - - - - -
{sid, name}, {Phno, sec}, - - - - - - - }

Candidate key :→ Minimal superkey is called Candidate key.

{{sid}, {Phno.}, {Adhno}}

DBA

Primary key :→ Only one Candidate key can be primary key for a relation.

{{sid}} {Unique Data} ⇒ Not Null.

(i) —

Alternate Key / Secondary key :→ {{Phno}, {Adhno}}.

Composite key :→ {{sid, name}, {Phno, sec}, {name, Roll, class, sec}}

Foreign Key :→

Student

sid	name	class	sub
S001	AA	12	Math Phy Comp.

P.K.

It may consist Duplicate value.

Teacher

Tid	name	sid
T001	XY	S001
T002	XZ	S001
T003	YZ	S001

P.K. ← Tid

sid → F.K.

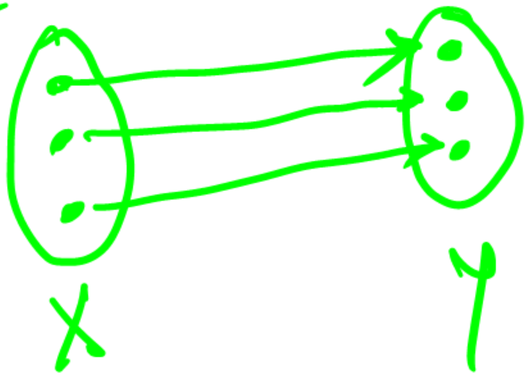
⇒ Foreign key may be more than one present in a table.

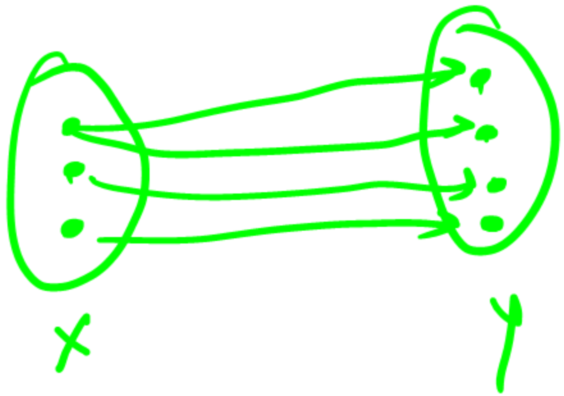
Relationship

- → 1 : 1 Relationship
- → 1 : M "
- → M : N "

Country Capital

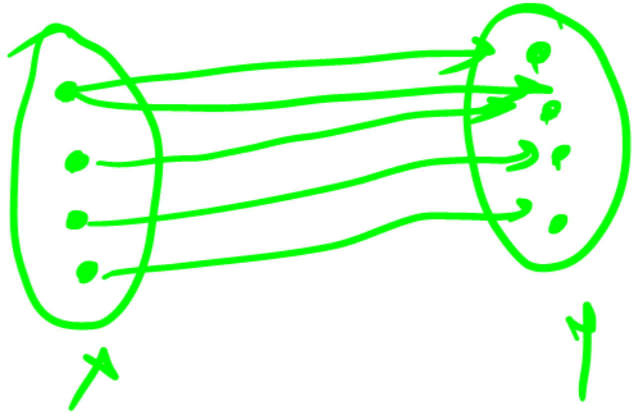
Dept HOD





Course Student

B.Tech



Product Customer

Next Topic :→ E-R Model with Diagram.